STA 445 S26 Assignment 5

2026-03-17

```r
library(tidyverse)
library(lubridate)
library(mosaicData)
```

Problem 1

Convert the following to date or date/time objects.

a. September 13, 2010.

```r
mdy("September 13, 2010.")
```

```
## [1] "2010-09-13"
```

b. Sept 13, 2010.

```r
mdy("Sept 13, 2010.")
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA
```

It doesn't work because September is usually abbreviated as sep, not sept.

c. Sep 13, 2010.

```r
mdy("Sep 13, 2010.")
```

```
## [1] "2010-09-13"
```

d. S 13, 2010. Comment on the month abbreviation needs.

```r
mdy("S 13, 2010.")
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA
```

The month abreviations need to be the first 3 letters of the month's name.

e. 07-Dec-1941.

```r
dmy("07-Dec-1941.")
```

```
## [1] "1941-12-07"
```

f. 1-5-1998. Comment on why you might be wrong.

```r
mdy("1-5-1998.")
```

```
## [1] "1998-01-05"
```

1 and 5 could both be either months or days, so it could be January 5 or May 1.

g. 21-5-1998. Comment on why you know you are correct.

```r
dmy("21-5-1998.")
```

```
## [1] "1998-05-21"
```

21 has to be the day because there is no 21st month.

h. 2020-May-5 10:30 am

```r
ymd_hm("2020-May-5 10:30 am")
```

```
## [1] "2020-05-05 10:30:00 UTC"
```

i. 2020-May-5 10:30 am PDT (ex Seattle)

```r
ymd_hm("2020-May-5 10:30 am", tz = "US/Pacific")
```

```
## [1] "2020-05-05 10:30:00 PDT"
```

j. 2020-May-5 10:30 am AST (ex Puerto Rico)

```r
ymd_hm("2020-May-5 10:30 am", tz = "America/Puerto_Rico")
```

```
## [1] "2020-05-05 10:30:00 AST"
```

Problem 2

Using just your date of birth (ex Sep 7, 1998) and today's date calculate the following:

a. Calculate the date of your 64th birthday.

```
birthday <- mdy("11-18-2005")
birthday + years(64)
```

```
## [1] "2069-11-18"
```

b. Calculate your current age (in years).

```
age <- interval(birthday, today()) %>%
  as.period() %>%
  year()
age
```

```
## [1] 20
```

c. Using your result in part (b), calculate the date of your next birthday.

```
next.birthday <- birthday + years(age + 1)
next.birthday
```

```
## [1] "2026-11-18"
```

d. The number of *days* until your next birthday.

```
interval(today(), next.birthday) %>%
  as.period(, unit = "days")
```

```
## [1] "246d 0H 0M 0S"
```

e. The number of *months* and *days* until your next birthday.

```
interval(today(), next.birthday) %>%
  as.period(unit = "months")
```

```
## [1] "8m 1d 0H 0M 0S"
```

Problem 3

Suppose you have arranged for a phone call to be at 3 pm on May 8, 2015 at Arizona time. However, the recipient will be in Auckland, NZ. What time will it be there?

```r
mdy_hm("5-8-2015 3:00 PM", tz = "US/Arizona") %>%
  with_tz(tz = "NZ")
```

```
## [1] "2015-05-09 10:00:00 NZST"
```

Problem 4

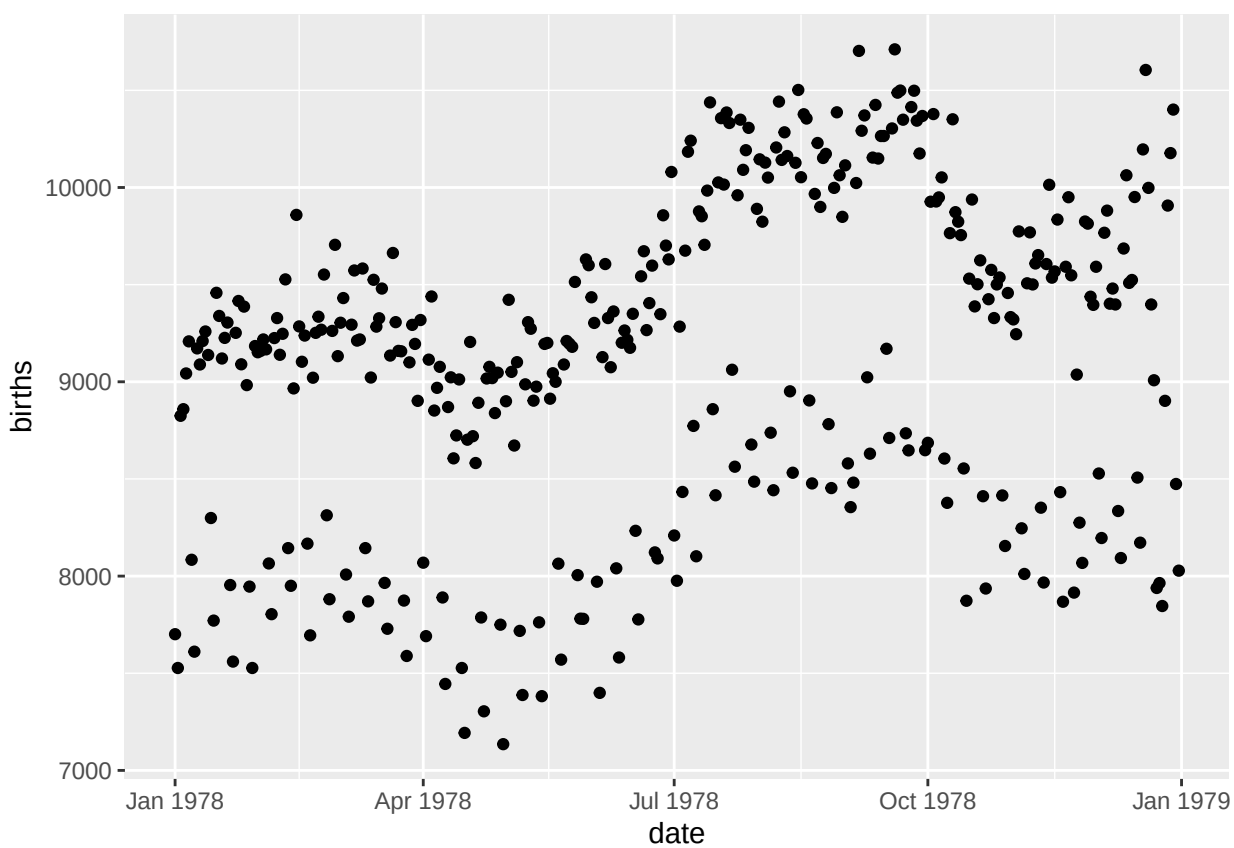
It turns out there is some interesting periodicity regarding the number of births on particular days of the year.

a. Using the `mosaicData` package, load the data set `Births78` which records the number of children born on each day in the United States in 1978. Because this problem is intended to show how to calculate the information using the `date`, remove all the columns *except* `date` and `births`.

```r
births <- select(Births78, date, births)
```

b. Graph the number of `births` vs the `date` with date on the x-axis. What stands out to you? Why do you think we have this trend?

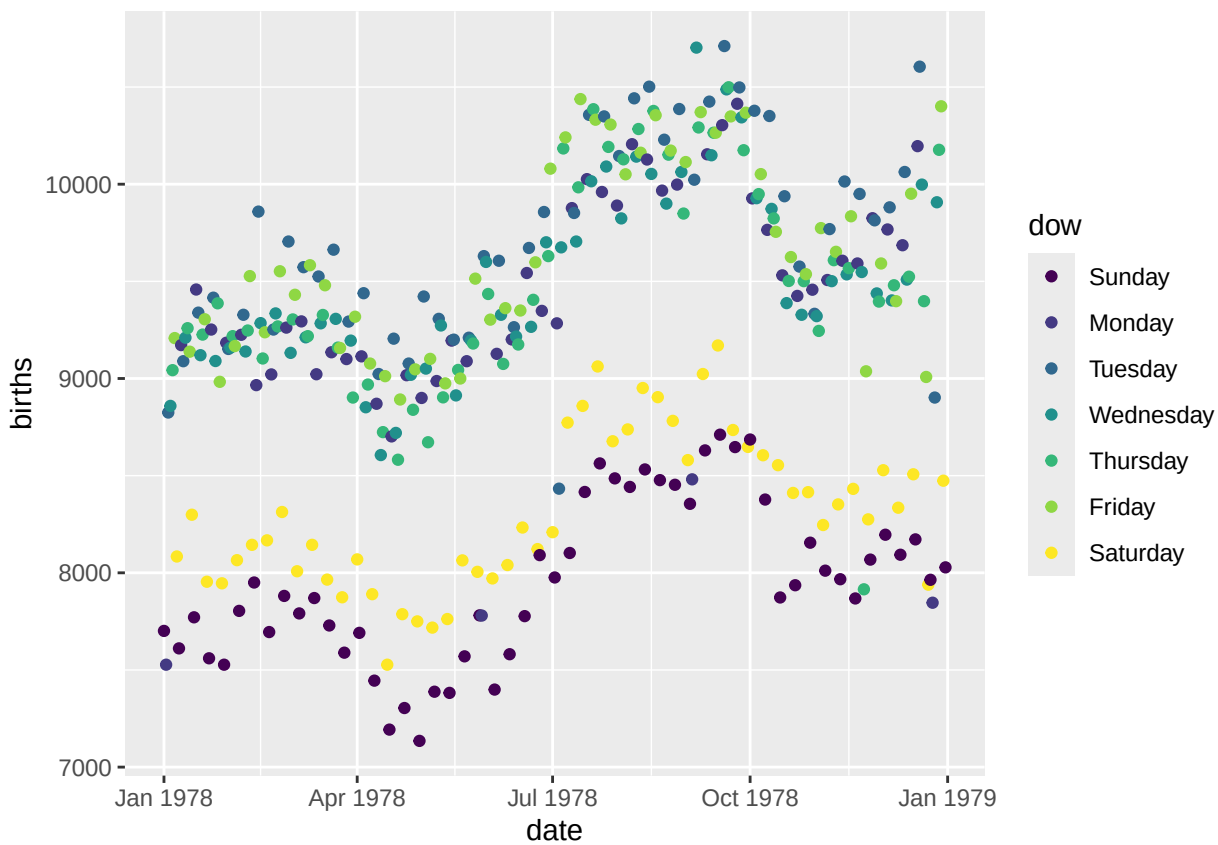
```r
ggplot(births, aes(date, births)) +
  geom_point(stat = "identity")
```



It looks like there are 2 separate trend lines, so it's probably split into 2 groups based on day of the week. It looks like the lower group has less dots than the higher group so I would guess it's the weekends. Maybe people are doing less stuff on the weekends, so they move around less and provoke birth less.

c. To test your assumption, we need to figure out the what day of the week each observation is. Use `dplyr::mutate` to add a new column named `dow` that is the day of the week (Monday, Tuesday, etc). This calculation will involve some function in the `lubridate` package and the `date` column.

```
births <- births %>%
  mutate(dow = wday(date, label = TRUE, abbr = FALSE))
```

d. Plot the data with the point color being determined by the day of the week variable.

```
ggplot(births, aes(date, births, color = dow)) +
  geom_point(stat = "identity")
```